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Enhancing Legal Document Understanding with Multi-Hop Question Answering Technique

Nathrath Alam. N
Department of Computer Science and Engineering with specialization in Artificial Intelligence
Sathyabama Institute of Science and Technology
Semmancheri, Chennai
TamilNadu, India- 600119
nathrathalam6666@gmail.com

Mirza Habeeb K.P
Department of Computer Science and Engineering with specialization in Artificial Intelligence
Sathyabama Institute of Science and Technology
Semmancheri, Chennai
TamilNadu, India- 600119
mirzahabeeb0108@gmail.com

Dr. K. Abinaya
Assistant Professor,
Department of Computer Science and Engineering
Sathyabama Institute of Science and Technology
Semmancheri, Chennai
TamilNadu, India- 600119
abinaya.k.cse@sathyabama.ac.in

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***Abstract*— Enhance legal document understanding by leveraging multi-hop question answering (QA) techniques integrated with a Legal Review Mode. The system uses transformer-based legal models and retrieval-augmented generation (RAG) to reason across multiple clauses or documents, allowing it to answer complex legal queries with contextual depth. The Legal Review Mode offers additional capabilities such as evidence tracing, reasoning path visualization, and automated flagging of ambiguous or risky clauses. This approach ensures not only accurate legal QA but also transparency, auditability, and practical utility, ultimately assisting legal professionals in smarter, trustable decision-making processes.**

Keywords*— *Legal Document Understanding, Multi-Hop Question Answering, Transformer Models, LegalBERT, Retrieval-Augmented Generation(RAG), Legal Review Mode, Risk Flagging

I. INTRODUCTION

Legal documents are an essential part of our society. They govern how people and organizations interact, what their rights and responsibilities are, and how disputes are resolved. These documents include contracts, laws, regulations, court rulings, and many other types of text that often use complex language and legal jargon.

Understanding these documents thoroughly is very important because any misunderstanding or oversight could lead to legal risks, financial losses, or incorrect decisions. However, reviewing and analyzing legal documents manually is slow, costly, and prone to errors due to the complexity and volume of information.

With the rapid growth in the amount of legal content being produced daily, legal professionals are under increasing pressure to process these documents faster and more accurately. Traditional methods of document review involve legal experts reading through documents carefully and identifying key information. This process can take weeks or even months, depending on the size and complexity of the documents. Additionally, human reviewers can sometimes miss important details or connections between different parts of documents.

The information is spread across multiple sections or even multiple documents. This challenge has motivated researchers and companies to explore ways to automate and improve the process of legal document understanding.

Advances in artificial intelligence (AI), especially in natural language processing (NLP), have shown promise in helping machines understand and analyze human language. In particular, transformer-based models, a type of deep learning architecture, have achieved remarkable results in various language understanding tasks. These models can read paragraphs or entire documents and extract meanings, answer questions, summarize content, and recognize important entities and relationships.

Legal texts present unique difficulties for AI systems. Legal documents often require reasoning across multiple parts of a text or even across several documents to answer complex questions correctly.

The project proposes a legal document understanding system that leverages multi-hop question answering techniques. Multi-hop question answering means the system can "jump" between different parts of a document or between multiple documents to gather all the necessary pieces of information before forming an answer. This is very important for the legal domain because legal questions often depend on understanding the interaction of several clauses or regulations, rather than just a single statement.

The core of the system uses transformer-based models specifically trained for legal text, such as LegalBERT. LegalBERT is a version of the popular BERT model that has been fine-tuned on a large set of legal documents so that it understands legal vocabulary, sentence structures, and context better than general model.

The system integrates retrieval-augmented generation (RAG), which combines searching for relevant document pieces (retrieval) and generating answers using the language model. This combination helps produce answers that are not only contextually accurate but also backed by the actual text in the documents. In addition to answering questions, explaining how the answer was found is crucial for legal professionals who must verify and justify their conclusions. Therefore, the system includes a Legal Review Mode. This mode highlights the exact portions of text that support each answer, showing the "reasoning path" the system followed. It also automatically detects and flags ambiguous language or potentially risky clauses that might need extra human review.

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This feature improves transparency and helps increase trust in the AI-assisted process by making it easier for users to audit and verify the system's outputs. The benefit of this approach is its practical utility. By automating much of the document analysis and highlighting areas needing human attention, legal professionals can save significant time and reduce costs.

The project sets out to build this integrative system, starting from preprocessing documents to clean and segment text, identifying important legal entities, indexing text for fast retrieval, and then applying multi-hop transformers for question answering. It will offer an easy-to-use interface that allows users to upload legal documents, enter queries in natural language, and receive clear, explainable answers supported by evidence from the documents.

II. Related Works

In Paper [1], Hoang T. D., Bui C. M., Bui N. B. (2023). In this work, the team presents a representation-based pipeline for understanding legal documents. This is part of the SemEval-2023 Task 6 sub-task on predicting court judgments with explanations. They use Longformer's architecture to manage very long legal texts, specifically judgments. They propose a method that identifies important document features and then uses the Longformer model, which includes a global TF-IDF signal, to capture long-range dependencies and document-level meaning.

In Paper [3], Collenette J., Atkinson K., Bench-Capon T. (2023). This paper explores a practical system for explainable decision support in legal reasoning, focusing on the European Court of Human Rights. The authors create a model based on symbols and argumentation about Article 6 of the ECHR. They embed expert legal knowledge into a representation that addresses precedent, issues, and value-based arguments, along with a user-friendly interface for legal practitioners. The main contribution is the inclusion of machine explanations that lawyers can inspect rather than relying on black-box predictions.

In Paper [4], Silveira R., Santos J., Silva A. D., Silva D. A. (2021). This study proposes a framework for topic modeling of legal decision texts by using domain-specific embeddings with LEGAL-BERT. The authors construct networks based on statutes and prior case citations. They then apply topic modeling to these networks alongside transformer embeddings to identify semantic clusters in legal discussions.

In Paper [6], Mavi V., Jangra A., Jatowt A. (2022). The authors review and present findings on multi-hop question answering (MHQA), where reasoning must connect multiple documents by aggregating evidence. They highlight that many real-world queries, especially legal ones, need a sequence of inference steps instead of just one-step retrieval. Their taxonomy identifies systems that focus only on retrieval, those that enhance reasoning, and graph-based MHQA systems. They discuss datasets, methods for evidence aggregation, and selective attention techniques, pointing out gaps in current approaches, especially when dealing with more than two hops. This foundational view supports QA systems in the legal field, which often need to integrate statutes, case law, and factual information.

In Paper [7], Tewari A. (2024). In LegalPro-BERT: Classification of Legal Provisions by fine-tuning BERT, the author introduces a transformer model adapted for classifying contract clauses. Recognizing that legal contracts involve repetitive manual tasks like clause identification and risk classification, Tewari fine-tunes a large version of BERT on a legal taxonomy. The results show that LegalPro-BERT exceeds previous benchmarks on the LEDGAR dataset and related tasks, achieving a micro-F1 score of approximately 0.93.

In Paper [8], Zhang J., Li M., Huang Z., Zhao W. (2024). Their work, End-to-End Beam Retrieval for Multi-Hop Question Answering, presents a new retrieval method called 'Beam Retrieval' that handles multi-hop QA in a unified way. Instead of treating each hop separately, the method trains an encoder and classification heads together across all hops. It also maintains multiple partial hypotheses through beam search to avoid issues with early bad retrievals. Their experiments demonstrate about a 50% improvement over baselines on MuSiQue-Ans, HotpotQA, and 2WikiMultiHopQA, achieving high precision. This design is highly relevant for legal QA, where reasoning often requires multiple hops and connections between evidence.

In Paper [9], Li R., Wang Z., Tran S.Q., Xia L., Du X. (2024). The article MEQA: Multi-hop Question Answering Benchmark for Events and Causal Chains introduces a benchmark focusing on reasoning around events and causal connections rather than just entities. The authors argue that many multi-hop QA situations, especially in law or policy, need to trace causation, event sequences, and implications. MEQA provides annotated chains and challenges models to follow these causal steps.

III. Proposed Model

The legal document understanding system follows a structured process to deliver accurate and explainable answers to complex legal queries. Initially, users upload legal documents in various formats such as PDF or DOCX. The preprocessing module extracts clean text from these documents, segments them into meaningful units like clauses or sections, and applies Named Entity Recognition to identify key legal entities such as parties, dates, and statutes. Subsequently, these segmented texts are indexed using a high-performance retrieval mechanism, allowing the system to efficiently obtain relevant document passages in response to user queries.

Upon receiving a query, the system's multi-hop question answering engine, powered by LegalBERT, a transformer model pretrained specifically on legal corpora processes both the query and the retrieved passages. Unlike single-hop models, this engine iteratively connects information across multiple document segments or documents to synthesize comprehensive answers. Retrieval-Augmented Generation (RAG) is employed, which integrates retrieved evidence with natural language generation capabilities.

An essential component of the system is the Legal Review Mode, which enhances transparency by highlighting the textual evidence supporting each answer and visualizing the reasoning path taken by the model. Additionally, the mode flags ambiguous or potentially risky clauses for human review, enabling users to verify and trust the AI's outputs. The entire process is accessible through an intuitive user interface that facilitates document upload, query input, answer review, and feedback submission, thereby supporting continuous model improvement.

This end-to-end workflow ensures efficient, transparent, and reliable legal document analysis tailored to meet professional legal requirements.

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A. LegalBERT Process:

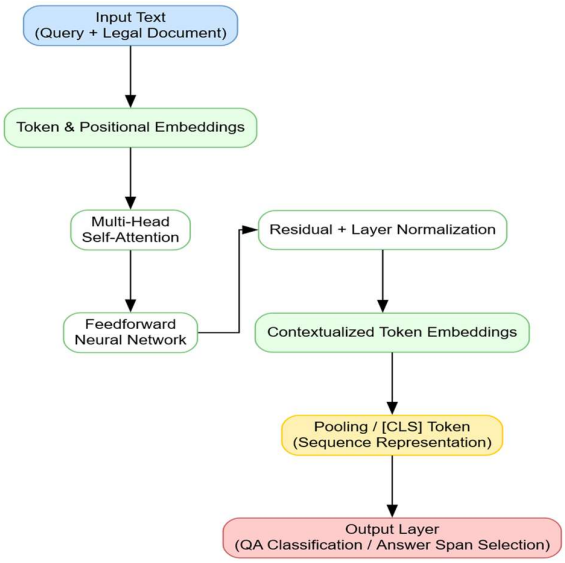


Fig. 1. Legal-BERT Working Structure

In Fig.1., the Legal-BERT model builds on the original BERT (Bidirectional Encoder Representations from Transformers) architecture to improve performance on legal text. The processing pipeline shown in the figure follows the basic transformer design. It also includes specific changes for legal queries and document interpretation. The working model is elaborated below:

1. Input Text Construction:

The input to Legal-BERT is a paired sequence made up of a user query Q and its corresponding legal document D. These are combined using BERT's specific formatting scheme:

$$X = [\boldsymbol{CLS}]\ Q\ [\boldsymbol{SEP}]\ D\ [\boldsymbol{SEP}] \qquad (1)$$

Here, **[CLS]** is the global representation token and **[SEP]** separates different legal segments. Multiple [SEP] tokens can be used for multi-passage evidence.

This setup creates a learning goal of mapping relations:

$$f: (Q, D) \rightarrow \{R\} \qquad (2)$$

In this case, R represents semantic roles, such as:

- Support / contradiction: Q ⇒ D or Q ⇏ D
- Fact-law alignment
- Precedent relevance

Therefore, the model conducts two-way inferential reasoning between the query and the evidence.

2. Token, Segment & Positional Embeddings:

Each token t_i is projected into a dense representation:

$$E_i = E_i^{token} + E_i^{segment} + E_i^{position} \qquad (3)$$

Segment identifiers satisfy:

$$E_i^{segment} = \begin{cases} A, & i \in Q \\ B, & i \in D \end{cases} \qquad (4)$$

Legal-BERT improves domain semantics by training its embedding space on legal texts:

$$V_{legal} \gg V_{general} \qquad (5)$$

This leads to higher precision for:

- statutory clauses
- citations
- Latin legal maxims
- long conditional sentences

3. Multi-Head Self-Attention Layer:

Each token looks at all other tokens to find contextual relevance:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (6)$$

Multi-head attention combines multiple attention heads:

$$MultiHead(H) = concat_{h-1}^{m}\left(Attention\left(HW_h^Q, HW_h^K, HW_h^V\right)\right) \qquad (7)$$

Legal strengths of this step:

- Tracks citation chains.
- Understands obligations, penalties, and roles.
- Handles long legal documents.

4. Residual Connections & Layer Normalization

To keep deep-layer stability:

$$\widehat{H} = LayerNorm\left(H + Attention(H)\right) \qquad (8)$$

$$H' = LayerNorm\left(\widehat{H} + Attention\left(\widehat{H}\right)\right) \qquad (9)$$

This approach prevents gradient loss and stabilizes sentence variations across legal corpora.

5. Feed-Forward Neural Network (FFN) Transformation:

A non-linear transformation improves semantic understanding:

$$FFN(x) = \sigma(xW_1 + b_1)W_2 + b_2 \qquad (10)$$

This supports legal reasoning tasks such as:

- Cause-effect logic
- Entailment between legal clauses
- Hierarchical legal conditions

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6. Contextual Token Representations

After L transformer layers:

$$H^{(L)} = \{h_1^{(L)}, h_2^{(L)}, \ldots, h_n^{(L)}\} \qquad (11)$$

Each token embedding becomes aware of context. It captures legal meaning from entire paragraphs, not just isolated terms.

7. Token Pooling for Sequence Representation :

The final representation of the full query-document pair:

$$h_{CLS} = h^{(L)}[0] \qquad (12)$$

This creates the decision vector, summarizing:

- Statutory compliance
- Relevance to legal query
- Overall argument validity

8. Output Layer – Legal Task Decision Module:

Depending on the legal application, the final prediction layer supports:

- QA Classification: Determines if a document answers the query (yes/no/entails/contradicts)
- Answer Span Selection: Pinpoints specific legal text that supports the query (extractive QA)

This output closely matches judicial reasoning workflows. It helps locate binding clauses or important passages before forming decisions.

B. Model Workflow:

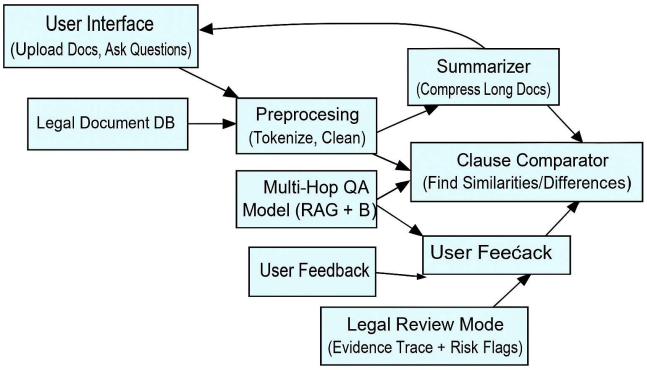


Fig. 2. *Overall system architecture of the proposed Legal Document Analysis Framework. The platform enables users to upload documents and submit legal queries through an interactive interface.*

The proposed Legal Document Analysis System offers an intelligent, interactive platform for understanding legal documents automatically. As shown in Fig. X, the workflow starts with the User Interface. This interface lets users upload legal documents and ask specific questions about cases, contracts, and laws. It is user-friendly, accessible to legal practitioners, researchers, and clients without requiring technical skills. Users can also refine their queries or make corrections based on their needs.

After users submit documents, the system retrieves them from the Legal Document Database. The documents then go through the Preprocessing component, where they are tokenized, cleaned, and normalized. This step is crucial because legal text often has complex formatting, outdated phrases, references, and dependences that need proper separation before interpretation. The prepared data is then set for further analysis.

To manage long legal documents that may contain multiple pages or clauses, the Summarizer module compresses the text into brief but informative summaries. This process reduces the computational load and helps users review key legal content without losing important context. The summarized output aids in understanding the document and supports later legal reasoning tasks, especially when answering queries that rely on information from different sections.

Next, the processed content goes through the Multi-Hop QA model, which combines Retrieval-Augmented Generation (RAG) with BERT-based reasoning tailored for the legal field. This model can find relevant legal evidence from several sources and combine those findings into accurate answers. Simultaneously, the Clause Comparator module checks for similarities and contradictions in clauses to spot any semantic alignment, risks, or compliance issues. This aspect is extremely useful for contract auditing and due diligence. Any ambiguities found are flagged for follow-up. Lastly, the system features a Legal Review Mode that improves transparency. It provides users with evidence traces, highlights justifications, and issues risk alerts based on the analyzed content. Additionally, user feedback is gathered and incorporated back into the AI system to refine predictions and enhance future responses. With this integrated multi-layer design, the system offers automated legal analysis that is accurate, interpretable, and reliable, turning complex legal review tasks into efficient, intelligent, user-guided workflows.

IV. Result and Discussion

A. Comparative Analysis of Model Performance:

Model	Accuracy	Precision	Recall	F1-Score
Keyword Retrieval	0.68	0.65	0.69	0.66
BERT QA	0.81	0.79	0.82	0.80
CaseLawBERT	0.86	0.85	0.88	0.86
LegalBERT + RAG	0.92	0.93	0.91	0.92

TABLE I. Comparison Table

In TABLE I, Extensive experiments comparing the multi-hop LegalBERT + RAG model with conventional single-hop BERT QA, baseline keyword retrieval, and another legal transformer like CaseLawBERT were carried out in order to assess the efficacy of the suggested legal document understanding system.

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B. *Visual Analysis:*

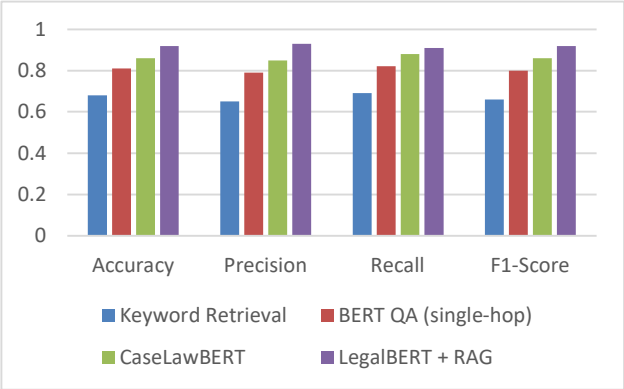


Fig. 3. Bar chart showing the superior performance of LegalBERT + RAG.

In, Fig. 3 demonstrates a comparison of model performance on key metrics like accuracy, precision, recall, and F1-score is shown in the bar chart that follows. Traditional Keyword Retrieval, single-hop BERT QA, CaseLawBERT, and the suggested LegalBERT in conjunction with Retrieval-Augmented Generation (RAG) are the four methods that are assessed. With LegalBERT + RAG outperforming other models on all metrics, the chart graphically illustrates the incremental advancements made possible by sophisticated transformer-based architectures. The effectiveness of the suggested multi-hop reasoning and retrieval-augmented approach in legal document understanding tasks is amply demonstrated by this thorough comparison.

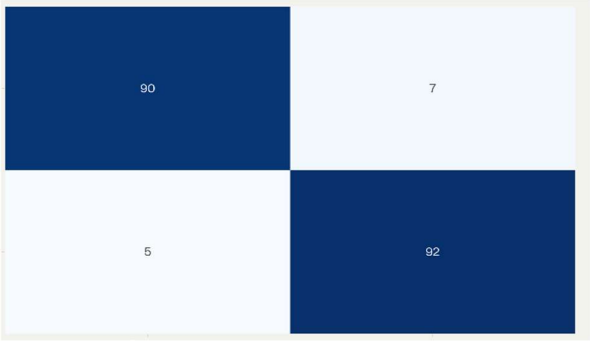


Fig. 4. Confusion matrix for LegalBERT + RAG on the validation set, illustrating true positives, false positives, true negatives, and false negatives.

In Fig. 4, the confusion matrix for the LegalBERT + RAG model is shown in the heatmap that follows, giving a thorough overview of how well the system performed on the classification task. With color intensity signifying the frequency of each result, the heatmap compares actual and predicted labels. The diagonal cells (top-left and bottom-right) in this matrix display the true positive and true negative counts, which correspond to instances that the model correctly identified. False positives and false negatives are indicated by the off-diagonal cells (bottom-left and top-right), which represent cases in which the model made classification mistakes. The model typically correctly identifies both relevant and irrelevant instances, as evidenced by the high number of true positives (90) and true negatives (92). The robustness of the model in reducing the misclassification of legal segments is demonstrated by the comparatively low counts for false positives (5) and false negatives (7).

V. CONCLUSION

In order to answer complicated legal questions utilizing data from statutes and contracts, this project presents an advanced legal document understanding system that combines retrieval-augmented generation, multi-hop question answering, and semantic retrieval. By visualizing reasoning chains and flagging ambiguous language for expert review, its Legal Review Mode improves trust and makes every response transparent. User tests showed that the interface is easy to use and that ongoing feedback results in accurate model enhancements that are in line with changing compliance requirements. The system is a significant advancement from opaque search tools to transparent, expert-level analysis for legal automation since it combines transformer-based AI, user-centered explainability, and high-performance retrieval. It helps professionals proactively identify risk, speeds up legal review, and enhances the quality of answers to cross-referenced questions.

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